Week 03 Face-to-Face Class Outline

Topic: Course Project Tutorial

1. Reiterate the timeline and goal of this course project (refer to Slide #20, #21 of Week 01)
2. Technical tips
    a. You may choose any application
    b. Choose an IoT platform (e.g., Android, iOS, Arduino, etc)
    c. You may use a cloud platform or a workstation computer to simulate the cloud server
        i. Some suggestions on how to use the free trial of Google Cloud
    d. You may choose any programming language
3. Marking criteria (refer to the rest of this document)
4. Demo of a completed course project

# Tasks (100 marks)

- Local inference — 40 marks
  - Collect user input (10)
  - Infer locally and display result (15)
  - Run on emulated/physical IoT device (15)
- Cloud inference — 30 marks
  - Run inference in cloud virtual machine (10)
  - Communicate btw IoT device & cloud (20)
- Advanced tasks — 30 marks
  - Train your own model (10)
  - Support multiple concurrent users (10)
  - Support online model updating (10)

# Marking Criteria (1)

- Collect user input (10)
  - **IF** load input from storage (5)
  - **ELSE** collect real-time input by touch screen, microphone, camera, built-in sensors (10)

# Marking Criteria (2)

- Infer locally and display result (15)
  - **IF** your app offloads the execution to cloud, (10)
  - **ELSE** run heuristic algorithm (not neural network) on the mobile app (5)
  - Display the inference result in real time by screen or synthetic voice (5)

# Marking Criteria (3)

- Run on emulated/physical IoT device (15)
  - **IF** the program runs natively on a desktop or laptop computer (5)
  - **ELSE** the program runs on an emulated or a physical IoT device (15)
    - iOS emulator provided by Xcode
    - Android emulator
    - Real smartphone
    - Raspberry Pi + add-on sensors (camera, microphone, etc)

# Marking Criteria (4)

- Run inference in cloud virtual machine (10)
  - **IF** deploy server program on a cloud virtual machine, e.g., Azure. (10)
  - **ELSE** deploy server program on your own computer (10)

# Marking Criteria (5)

- Communication btw IoT device and cloud (20)
  - **IF** HTML-based communications (5)
  - **ELSE**
    - Send the user input from the mobile app to the cloud for inference (10)
    - Send the inference result from the cloud to the mobile app (10)

# Marking Criteria (6)

- Train your own model (10)
  - **IF** train the used ML model by yourself (10)
    - Training program should be in the code package
    - Training results (e.g., accuracy) in a readme file
    - The training data can be any publicly available dataset
  - **ELIF** use downloaded pre-trained model (5)
  - **ELSE** use heuristic algorithm (not neural network) (2)

# Marking Criteria (7)

- Support multiple concurrent users (10)
  - Demonstrate multiple IoT devices can use the cloud service simultaneously

# Marking Criteria (8)

- Online model updating (10)
  - Demonstrate that the model in the cloud can be updated at run time using newly collected user data
    - User input and the corresponding label are transmitted to cloud
    - The cloud retrains the model